Arash Nabbi

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Education

PhD. Medical Science, Cancer Biology — September 2010-January 2017
University of Calgary, Cumming school of Medicine, Calgary, AB
Thesis: "The role of the ING3 epigenetic regulator in prostate cancer"

PharmD. Pharmacy — January 2004-December 2009
Azad University of pharmaceutical sciences, Tehran, Iran
Thesis: "The study of Noscapine-induced apoptosis in Lymphoblasts derived from patients with leukemia"

Research Experience

Scientific Associate I. Princess Margaret Cancer Centre — From August 2022
University Health Network, Toronto, ON
Mentor: Dr. Trevor Pugh

Post-doctoral Fellow. Princess Margaret Cancer Centre — September 2017- 2022
University Health Network, Toronto, ON
Mentor: Dr. Trevor Pugh

PhD Student. Arnie Charbonneau Cancer Institute — September 2010-January 2017
University of Calgary, Calgary, AB
Supervisor: Dr. Karl Riabowol

PharmD Student. Razi Institute of Drug Research — May 2007-December 2009
Iran University of Medical Sciences, Tehran, Iran
Supervisor: Drs. Massoud Mahmoudian and Nahid Rahbar

Teaching Experience

Graduate Teaching Assistant. — September 2014-April 2016
Human Integrative Physiology. Bachelor of health sciences.
University of Calgary, Calgary, AB

Clinical Experience

Clinical pharmacist. Shoppers drug mart — From May 2021
Toronto, ON

Pharmacy Intern. Sandstone enhanced care solutions — January 2017-August 2017
Calgary, AB

Publications

Manuscripts in revision/preparation

1. **Nabbi, A.,** Beck, P., Delaidelli, A., Oldridge, D. A., Sudhaman, S., Zhu, K., Yang, S. Y. C., Mulder, D. T., Bruce, J. P., Paulson, J. N., Raman, P., Zhu, Y., Resnick, A. C., Sorensen, P. H., Sill, M., Brabetz, S., Lambo, S., Malkin, D., Johann, P. D., Kool, M., Jones, D. T. W., Pfister, S. M., Jäger, N. & Pugh, T. J. Transcriptional immunogenomic analysis reveals distinct immunological clusters in pediatric nervous system tumours. *In review. Genome Medicine.*
2. **Nabbi, A.,** Pedersen S., Wellum J., Barrett D., Pugh, T. J. Effects of chemoradiation on circulating T-cell populations and repertoire in pediatric cancer patients. *In preparation.*

Peer reviewed papers (*: co-first author)

3. **Nabbi A**, Danesh A, Espin-Garcia O, Pedersen S, Wellum J, Fu LH, Geoerger B, Marshall LV, Trippett T, Rossato G, Pugh TJ, Hutchinson KE. Multimodal immunogenomic biomarker analysis of tumours from pediatric patients enrolled to a phase 1-2 study of single-agent atezolizumab. *Accepted. Nature Cancer.*
4. **Nabbi, A.*,** Beck, P.*, Delaidelli, A., Oldridge, D. A., Sudhaman, S., Zhu, K., Yang, S. Y. C., Mulder, D. T., Bruce, J. P., Paulson, J. N., Raman, P., Zhu, Y., Resnick, A. C., Sorensen, P. H., Sill, M., Brabetz,

S., Lambo, S., Malkin, D., Johann, P. D., Kool, M., Jones, D. T. W., Pfister, S. M., Jäger, N. & Pugh, T. J. Transcriptional immunogenomic analysis reveals distinct immunological clusters in pediatric nervous system tumours. 2022.09.20.508719 Preprint at https://doi.org/10.1101/2022.09.20.508719 (2022)

5. Beck, P., Selle, B., Madenach, L., Jones, D. T. W., Vokuhl, C., Gopisetty, A., **Nabbi, A**., Brecht, I. B., Ebinger, M., Wegert, J., Graf, N., Gessler, M., Pfister, S. M. & Jäger, N. The genomic landscape of pediatric renal cell carcinomas. iScience 25, (2022).
6. Das A, Sudhaman S, Morgenstern D, Coblentz A, Chung J, Stone SC, Alsafwani N, Liu ZA, Abu Al Karsaneh O, Soleimani S, Ladany H, Chen D, Zatzman M, Cabric V, Nobre L, Bianchi V, Edwards M, Sambira Nahum LC, Ercan AB, **Nabbi A**, et al. Genomic predictors of response to PD-1 inhibition in children with germline DNA replication repair deficiency. Nat Med. 2022;28:125–35.
7. Fink D, Yau T, **Nabbi A**, Wagner B, Wagner C, Hu SM, et al. Loss of Ing3 Expression Results in Growth Retardation and Embryonic Death. Cancers (Basel). 2019 Dec 29;12(1).
8. Dantas A, Al Shueili B, Yang Y, **Nabbi A**, Fink D, Riabowol K. Biological Functions of the ING Proteins. Cancers (Basel). 2019 Nov 19;11(11).
9. McClurg UL, Chit NCTH, Azizyan M, Edwards J, **Nabbi A**, Riabowol KT, Nakjang S, McCracken SR, Robson CN. Molecular mechanism of the TP53-MDM2-AR-AKT signalling network regulation by USP12. Oncogene. 2018;37(34):4679–91.
10. McClurg UL*, **Nabbi A***, Ricordel C*, Korolchuk S, McCracken S, Heer R, Wilson L, Butler LM, Irving-Hooper BK, Pedeux R, Robson CN, Riabowol KT, Binda O. Human ex vivo prostate tissue model system identifies ING3 as an oncoprotein. Br J Cancer. 2018 06;118(5):713–26.
11. Wang F, Wang AY, Chesnelong C, Yang Y, **Nabbi A**, Thalappilly S, Alekseev V, Riabowol K. ING5 activity in self-renewal of glioblastoma stem cells via calcium and follicle stimulating hormone pathways. Oncogene. 2018 Jan 18;37(3):286–301.
12. **Nabbi A**, McClurg UL, Thalappilly S, Almami A, Mobahat M, Bismar TA, Binda O, Riabowol KT. ING3 promotes prostate cancer growth by activating the androgen receptor. BMC Med. 2017 16;15(1):103.
13. Almami A, Hegazy SA, **Nabbi A**, Alshalalfa M, Salman A, Abou-Ouf H, Riabowol K, Bismar TA. ING3 is associated with increased cell invasion and lethal outcome in ERG-negative prostate cancer patients. Tumour Biol. 2016 Jul;37(7):9731–8.
14. McClurg UL, Harle VJ, **Nabbi A**, Batalha-Pereira A, Walker S, Coffey K, Gaughan L, McCracken SRC, Robson CN. Ubiquitin-specific protease 12 interacting partners Uaf-1 and WDR20 are potential therapeutic targets in prostate cancer. Oncotarget. 2015 Nov 10;6(35):37724–36.
15. Thakur S, **Nabbi A**, Klimowicz A, Riabowol K. Stromal ING1 expression induces a secretory phenotype and correlates with breast cancer patient survival. Mol Cancer. 2015 Aug 27;14:164.
16. **Nabbi A**, Almami A, Thakur S, Suzuki K, Boland D, Bismar TA, Riabowol K. ING3 protein expression profiling in normal human tissues suggest its role in cellular growth and self-renewal. Eur J Cell Biol. 2015 May;94(5):214–22.

Review articles

1. Jones DTW, Banito A, Grünewald TGP, Haber M, Jäger N, Kool M, Milde T, Molenaar JJ, **Nabbi A**, Pugh TJ, Schleiermacher G, Smith MA, Westermann F, Pfister SM. Molecular characteristics and therapeutic vulnerabilities across paediatric solid tumours. Nat Rev Cancer. 2019 Aug;19(8):420–38.
2. **Nabbi A**, Satpathy Sh, Riabowol K. Demethylating agents as epigenetic anticancer therapeutics. Curr Can Ther Rev. 2013; 9: 24-33.
3. Satpathy S, **Nabbi A**, Riabowol K. RegulatING chromatin regulators: post-translational modification of the ING family of epigenetic regulators. Biochem J. 2013 Mar 15;450(3):433–42.

Book chapters

1. **Nabbi A**, Riabowol K. Isolation of Nuclei. Cold Spring Harb Protoc. 2015 Aug 3; 2015(8):731–4.
2. **Nabbi A**, Riabowol K. Isolation of Pure Nuclei Using a Sucrose Method. Cold Spring Harb Protoc. 2015 Aug 3;2015(8):773–6.
3. **Nabbi A**, Riabowol K. Rapid Isolation of Nuclei from Cells In Vitro. Cold Spring Harb Protoc. 2015 Aug 3;2015(8):769–72.

Knowledge Dissemination

Invited talks and lectures

1. **Transcriptional analysis identifies distinct immune clusters in pediatric nervous system tumors.** The annual Children Brain Tumor Network (CBTN) scientific summit. October 2022

Washington, DC (Invited short talk)
https://www.youtube.com/watch?v=EauYUzwsXqw&ab_channel=Children%27sBrainTumorNetworkCBTN

2. **Transcriptional analysis reveals distinct microenvironmental subgroups across pediatric nervous system tumors.**
The International Symposium on Pediatric Neuro-Oncology. June 2022
Hamburg, Germany (Invited short talk)
3. **The immunogenomic landscape of pediatric primary solid tumors.** March 2021
Roche IMCORE pediatric workshop.
Virtual. (Invited short talk)
4. **The immunogenomic landscape of pediatric primary solid tumors.** September 2021
2019 AACR Advances in Pediatric Cancer Research.
Montreal, Quebec. (Invited short talk)
5. **The immunogenomic landscape of pediatric primary solid tumors.**
The annual Children Brain Tumor Network (CBTN) investigator meeting. May 2021
Virtual (Invited short talk)
https://www.youtube.com/watch?v=DDqc8mej_8M&ab_channel=Children%27sBrainTumorNetworkCBTN
6. **Immunogenomic characterization of pediatric tumor microenvironment**. December 2018
Hopp Children's Cancer Centre (KiTZ).
Heidelberg, Germany (Invited speaker)
7. **ING3: A novel co-activator of Androgen Receptor.** May 2017
Pharmaceutical outcomes program, BC Children's hospital research institute.
Vancouver, BC. (Invited speaker)
8. **Diuretics and the kidney**. March 2016
MDSC404 course. Calgary, AB. (University lecture)
9. **Nonrespiratory functions of the lungs**. January 2016
MDSC404 course. Calgary, AB. (University lecture)
10. **Biomarker discovery: Lessons learned from prostate cancer study**. May 2015
Lung Cancer research group meeting. Calgary, AB. (Invited speaker)

Poster presentations

1. **Transcriptional immunogenomic analysis to identify inflamed immune microenvironment across pediatric solid tumors.** November 2022
The Terry Fox Research Institute's 9th Scientific Meeting
Vancouver, BC
2. **Landscape of infiltrating immune repertoire in pediatric solid tumors.** April 2019
American Association of Cancer Research (AACR) 2019 annual meeting.
Atlanta, GA
3. **Immunogenomic characterization of pediatric solid tumors.** December 2018
Terry Fox Research Institute Ontario Node Research Symposium.
Toronto, ON
4. **The ING3 epigenetic regulator is a novel biomarker for prostate cancer.** November 2015
Canadian cancer research conference 2015.
Montreal, QC
5. **ING3 modulates Androgen Receptor activity in prostate cancer**. January 2015
Southern Alberta Cancer Research Institute (SACRI) Research Day.
Calgary, AB
6. **The effects of ING3 on prostate cancer cell survival and invasion**. July 2014
7. 23rd European Association for Cancer Research, Munich, Germany.
8. **Deciphering role of ING3 on androgen receptor activity in prostate cancer cells**. October 2013
Alberta Cancer Foundation (ACF) Conference,
Banff, AB
9. **Noscapine induces apoptosis on lymphoblasts derived from childhood acute lymphoblastic leukemia patients**. May 2007
The 7th International Symposium on Targeted Anticancer Therapies.
Amsterdam, Netherlands

Awards and recognitions

CBTN Pioneering Spirit Award winner	May 2021
AACR Scholar-in-Training Award (250 USD). Advances in Pediatric Cancer Research, Montreal, QC	September 2019
Princess Margaret Post-Doctoral Fellowship award (50,000 CAD). Princess Margaret Cancer Centre, Toronto, ON	January 2018
Queen Elizabeth II scholarship (15,000 CAD). Faculty of Graduate Studies, University of Calgary, Calgary, AB	September 2016
Queen Elizabeth II scholarship (15,000 CAD). Faculty of Graduate Studies, University of Calgary, Calgary, AB	September 2015
MDSC Graduate Program travel award (1,500 CAD) University of Calgary, Calgary, AB	June 2014
CIHR Training Program Travel Award-ACHRI (750 CAD) University of Calgary, Calgary, AB	June 2014
Program Specific Award (PSA) For first author publication (500 CAD) Medical Science Graduate Program, University of Calgary, Calgary, AB	January 2013
Program Specific Award (PSA) (2,550 CAD) Medical Science Graduate Program, University of Calgary, Calgary, AB	January 2012

Professional Certificates

Licensed under Ontario College of Pharmacists (#625769)	March 2018
Injectable Medication and Vaccine Administration Training Program	February 2017
Standard First Aid - CPR/AED Level C	March 2021
Pharmacy Examination Board of Canada certificate (#42298)	December 2016

Active Memberships

Society for Immunotherapy of Cancer (SITC)
Ontario College of Pharmacists. Member (#625769)
American Association of Cancer Research (AACR)

Skills

Technical skills
DNA library preparation, targeted capture sequencing, antibody validation, tissue culture, western blotting, qPCR, immunoprecipitation, immunofluorescence, molecular cloning, lentiviral preparation
Computer skills
R, Python, Bash, Adobe Illustrator, Jupyter notebooks, AWS, Docker
Language skills
English, German, Farsi

February 28, 2023